

AI-Powered Customer Sentiment Analysis for E-Commerce Websites

Project Overview

The AI-Powered Customer Sentiment Analysis for E-Commerce Websites project is designed to analyze customer reviews and automatically classify them into Positive, Neutral, and Negative sentiments using Artificial Intelligence, Machine Learning, and Deep Learning techniques. The system helps customers and businesses understand product feedback efficiently by converting large volumes of textual reviews into meaningful insights.

Objectives

- Analyze customer reviews from e-commerce platforms.
- Classify reviews into Positive, Neutral, and Negative sentiments.
- Generate sentiment scores and overall product ratings.
- Provide visual analysis through charts and graphs.
- Assist businesses in understanding customer satisfaction and product performance.

Features

- User Registration and Login System
- Product Review Analysis
- Real-Time Sentiment Prediction
- Sentiment Distribution Visualization
- Rating Generation Based on Reviews
- Interactive Dashboard
- Chatbot Assistance
- Report Generation

Technologies Used

Frontend: HTML, CSS, JavaScript

Backend: Python, Flask

Database: MongoDB Atlas

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras, Transformers (BERT)

Data Processing Libraries: Pandas, NumPy, NLTK, BeautifulSoup, Selenium

System Workflow

1. User enters a product URL or review text.
2. Reviews are collected through datasets or web scraping.
3. Text preprocessing is performed.
4. Features are extracted using TF-IDF.
5. Machine Learning and Deep Learning models analyze reviews.

6. Sentiments are classified as Positive, Neutral, or Negative.
7. Ratings and sentiment scores are generated.
8. Results are displayed through graphs and dashboards.

Machine Learning Models

Logistic Regression, Support Vector Machine (SVM), LSTM (Long Short-Term Memory), BiLSTM (Bidirectional LSTM), and BERT (Bidirectional Encoder Representations from Transformers).

Dataset

The project uses customer review datasets collected from E-commerce review datasets, product review datasets, and custom collected review data.

Installation

Clone Repository: `git clone https://github.com/yourusername/sentiment-analysis-project.git`
Create Virtual Environment: `python -m venv .venv`
Activate Environment (Windows): `.venv\Scripts\activate`
Install Dependencies: `pip install -r requirements.txt`
Run Application: `python app.py`

Project Structure

```
project/  
├── app.py  
├── requirements.txt  
├── models/  
├── datasets/  
├── prediction/  
├── static/  
├── templates/  
├── chatbot/  
├── reports/  
└── README.md
```

Expected Outcomes

Accurate sentiment classification, better understanding of customer opinions, improved product decision-making, and enhanced customer satisfaction analysis.

Future Scope

Multi-language sentiment analysis, voice review sentiment analysis, real-time social media sentiment monitoring, recommendation system integration, and advanced AI chatbot support.

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License

This project is developed for educational and research purposes.